

CS3700 - Assignment 1

State Election Database Management System

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1 Introduction

Elections are an essential part of democratic governance in India. State elections are conducted periodically to elect representatives to the legislative assemblies of each state. The election process involves multiple participants such as voters, candidates, political parties, and election officials.

The objective of this project is to design a structured system that can organize and manage information related to the election process. This includes maintaining records about political parties, candidates contesting in different constituencies, registered voters, polling booths, and election results.

2 Domain Description

This database is designed to manage information related to a state election conducted in a particular year. In a state election, the entire state is divided into several constituencies. Each **Constituency** represents a specific geographical area where eligible voters participate in the voting process to elect their representative to the state legislative assembly.

In every constituency, multiple **Candidates** contest the election. A candidate may represent a **Political party** or may contest independently without being associated with any party. Each candidate is allowed to contest from only one constituency in a particular election.

Candidates often organize campaign activities to interact with voters and promote their ideas and policies. These campaign events are conducted at specific venues within a constituency. Each **Campaign event** has a name, and it is organized by a particular candidate. These events allow candidates to communicate with voters and encourage public participation in the election process.

Registered **Voters** belonging to a constituency participate in the voting process by casting their vote for one of the candidates contesting in that constituency. Voting takes place at designated **Polling booths**. Each constituency contains multiple polling booths in order to manage voters efficiently and avoid overcrowding during the election process.

The entire election process is organized and supervised by the Election Commission. The Election Commission appoints **Election officers** who are responsible for conducting and monitoring the election in different constituencies. Each officer is assigned to a particular constituency and may supervise one or more polling booths or perform other administrative duties related to the election process.

After the voting process is completed, all votes cast in the polling booths are counted. Based on the vote counts, the candidate who receives the majority of votes in the constituency is declared the winner. The final **Results** are then officially announced for each constituency.

The purpose of this database is to systematically store and manage information related to constituencies, candidates, political parties, voters, polling booths, election officers, campaign events, votes cast during the election, and the final election results.

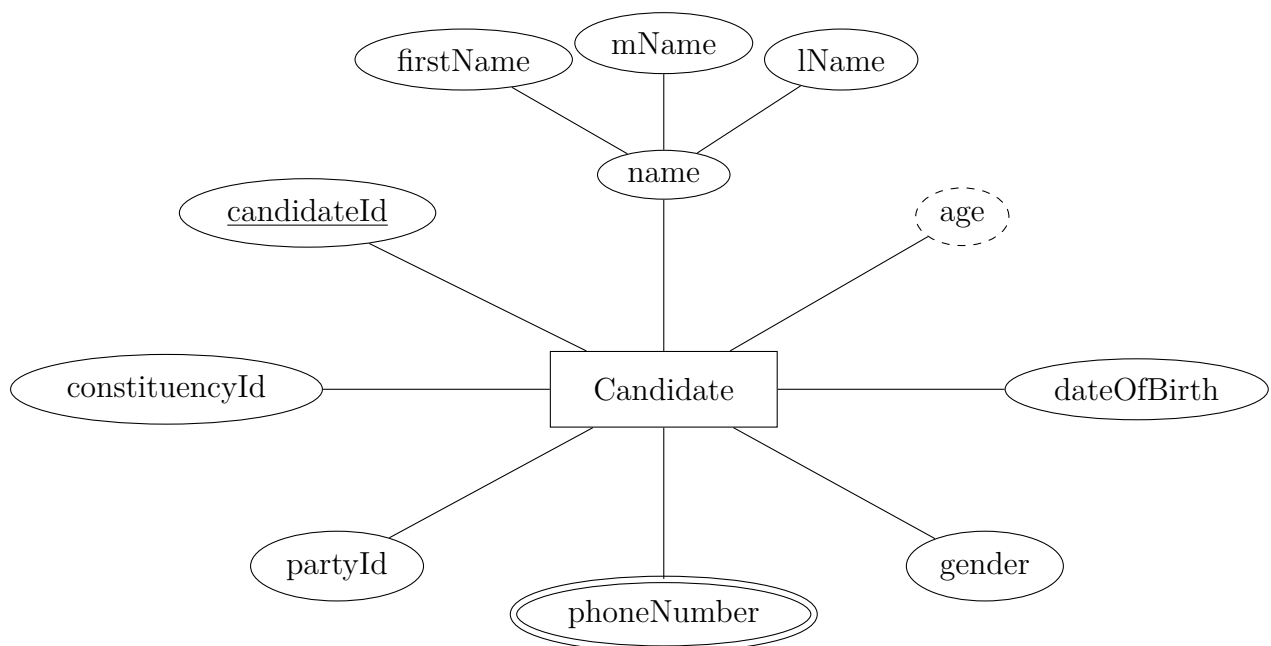
3 Entities

The election system involves several important components that store information required for managing elections. Some of the key components are described below.

3.1 Candidate

Represents individuals who contest in elections. A candidate may represent a political party or may contest independently. Each candidate contests from a single constituency and submits a nomination before the election. The system stores personal details, party affiliation, and nomination information of every candidate.

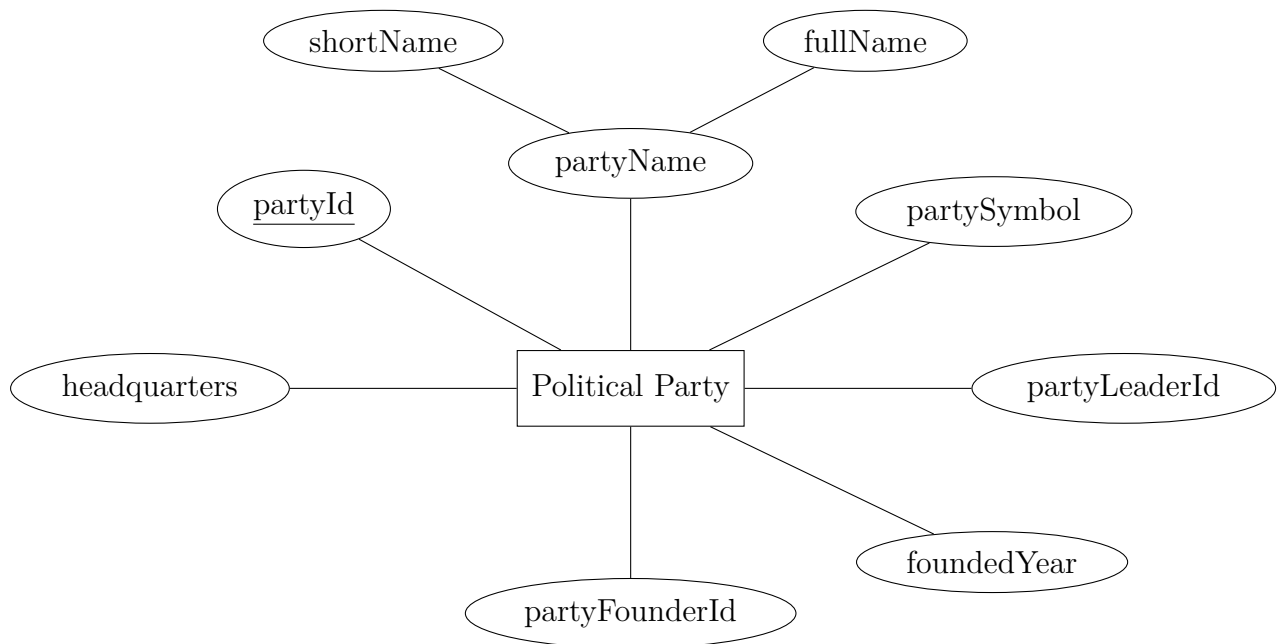
- **candidateId** : Uniquely identifies a candidate contesting in the election and acts as the primary key.
- **name** : A composite attribute consisting of first middle and last name.
- **dateOfBirth** : Stores the birth date of the candidate.
- **age** : A derived attribute that can be calculated from the date of birth.
- **gender** : Stores the gender of the candidate.
- **phoneNumber** : A multivalued attribute because a candidate may have multiple phone numbers.
- **partyId** : Identifies the political party the candidate belongs to. If the candidate is independent, this value is recorded as 0.
- **constituencyId** : Identifies the constituency that candidate is contesting in.



3.2 Political Party

Represents political organizations that participate in the election process by nominating candidates to contest from different constituencies. Each political party has a unique identity, a symbol used during elections, and leadership information. The database stores basic details about every political party involved in the election.

- **partyId** : Uniquely identifies each political party and acts as the primary key.
- **partyName** : Name of the political party. It is modeled as a composite attribute consisting of short name and full name.
- **partySymbol** : Represents the election symbol associated with the political party.
- **partyLeaderId** : Identifier of the current leader of the political party.
- **foundedYear** : The year in which the political party was established.
- **partyFounderId** : Identifier of the founder of the political party.
- **headquarters** : The main office location of the political party.

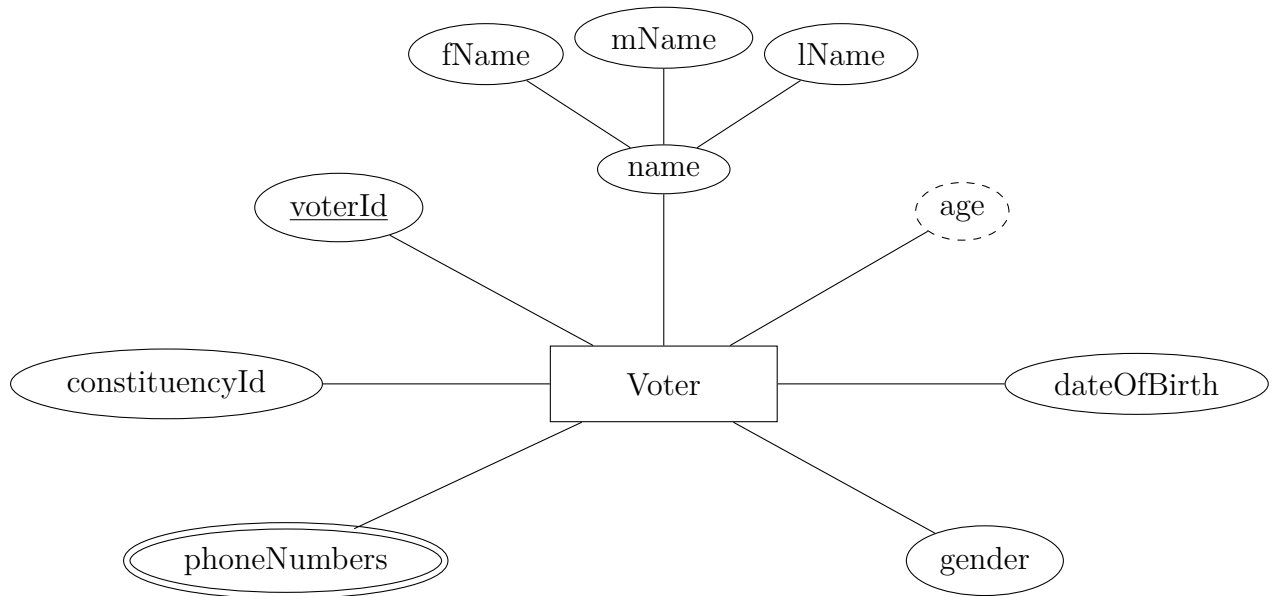


3.3 Voter

Represents citizens who are eligible and registered to vote in the election. Each voter belongs to a specific constituency and is identified by a unique voter identification number. The system stores personal information and contact details of voters to maintain accurate election records.

- **voterId** : Uniquely identifies each voter in the system and acts as the primary key.
- **name** : A composite attribute consisting of first middle and last name.

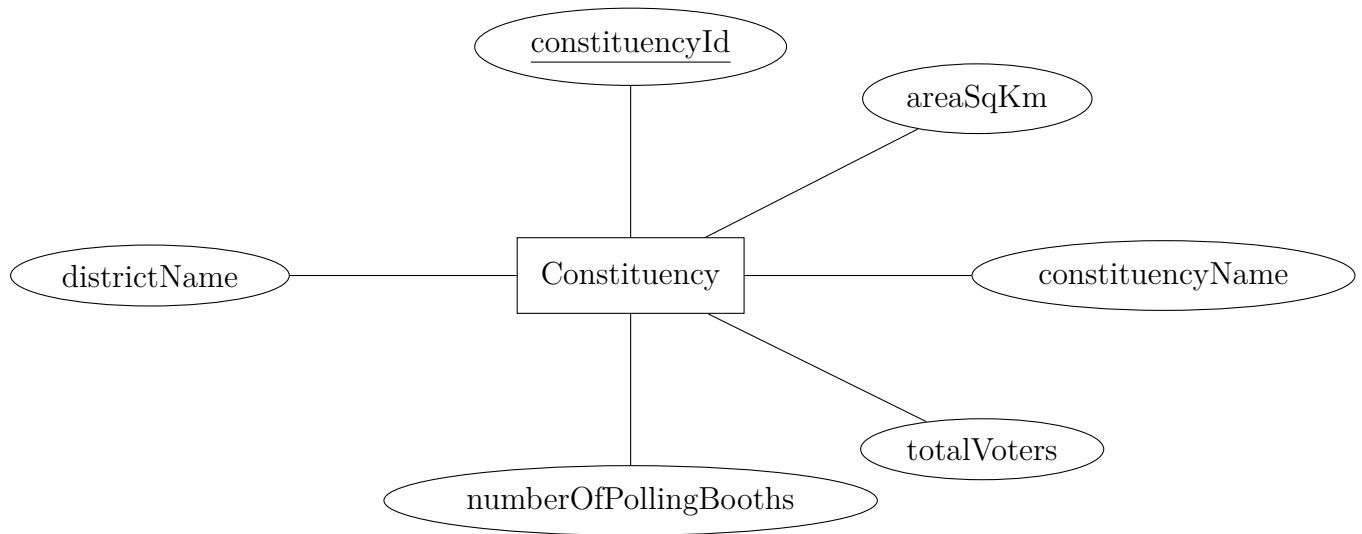
- **dateOfBirth** : Stores the birth date of the voter.
- **age** : A derived attribute calculated from the date of birth.
- **gender** : Stores the gender of the voter.
- **constituencyId** : Identifies the constituency to which the voter belongs.
- **phoneNumbers** : A multivalued attribute since a voter may have multiple contact numbers.



3.4 Constituency

Represents a geographical region within a state where voters participate in the election to elect their representative. A state is divided into multiple constituencies, and each constituency contains registered voters, candidates contesting the election, and several polling booths where voting takes place. The database stores information about each constituency to manage voters, polling booths, and election activities within that region.

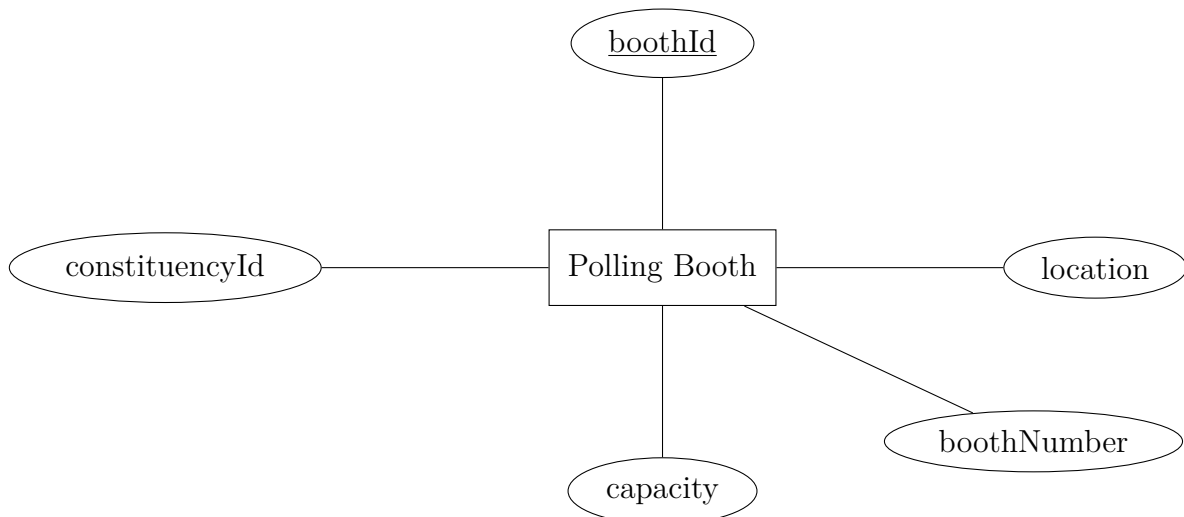
- **constituencyId** : Uniquely identifies a constituency and acts as the primary key.
- **constituencyName** : The official name of the constituency.
- **areaSqKm** : Represents the geographical area of the constituency measured in square kilometers.
- **totalVoters** : Stores the total number of registered voters in the constituency.
- **numberOfPollingBooths** : Represents the total number of polling booths present in the constituency.
- **districtName** : Indicates the district in which the constituency is located.



3.5 Polling Booth

Represents the physical location where voters cast their votes during the election. Each polling booth is located within a specific constituency and is used to manage voters efficiently during the voting process. Polling booths help distribute voters across multiple locations to reduce crowding and ensure that voting takes place smoothly and in an organized manner.

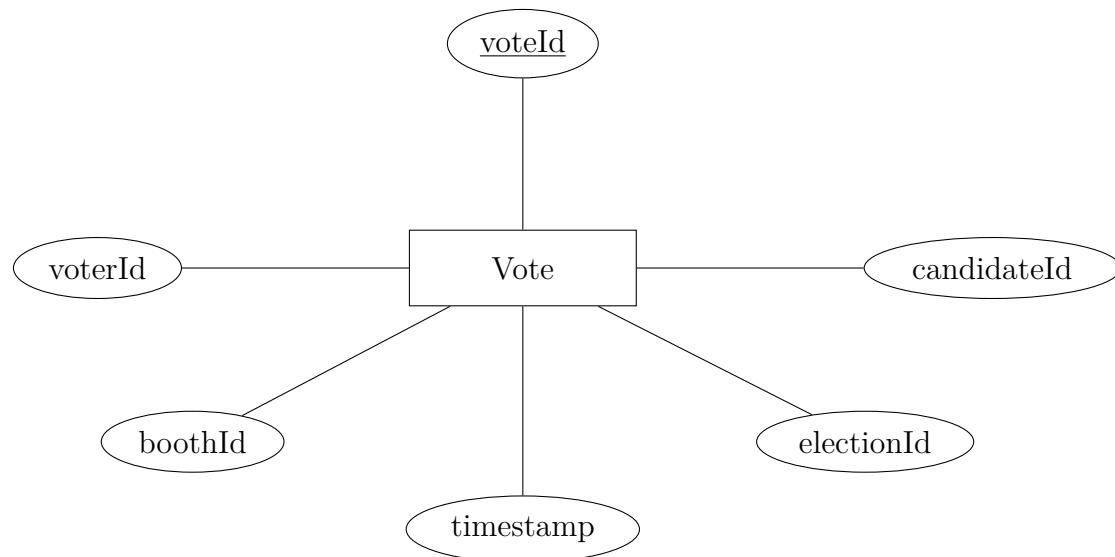
- **boothId** : Uniquely identifies each polling booth and acts as the primary key.
- **location** : Specifies the physical location or address where the polling booth is situated.
- **constituencyId** : Identifies the constituency in which the polling booth is located.
- **capacity** : Indicates the maximum number of voters that can be handled by the polling booth during the election.
- **boothNumber** : A unique booth number assigned within a constituency to help identify booths easily.



3.6 Vote

Represents the act of casting a vote by a registered voter during the election. Each vote is associated with a particular voter, the candidate they voted for, and the polling booth where the vote was cast. The database records every vote to ensure transparency and accurate counting of election results. The timestamp records the exact time at which the vote was cast.

- **voteId** : Uniquely identifies each vote cast in the election and acts as the primary key.
- **voterId** : Identifies the voter who cast the vote.
- **candidateId** : Identifies the candidate for whom the vote was cast.
- **boothId** : Identifies the polling booth where the vote was recorded.
- **electionId** : Identifies the election in which the vote was cast.
- **timestamp** : Stores the exact date and time when the vote was cast.

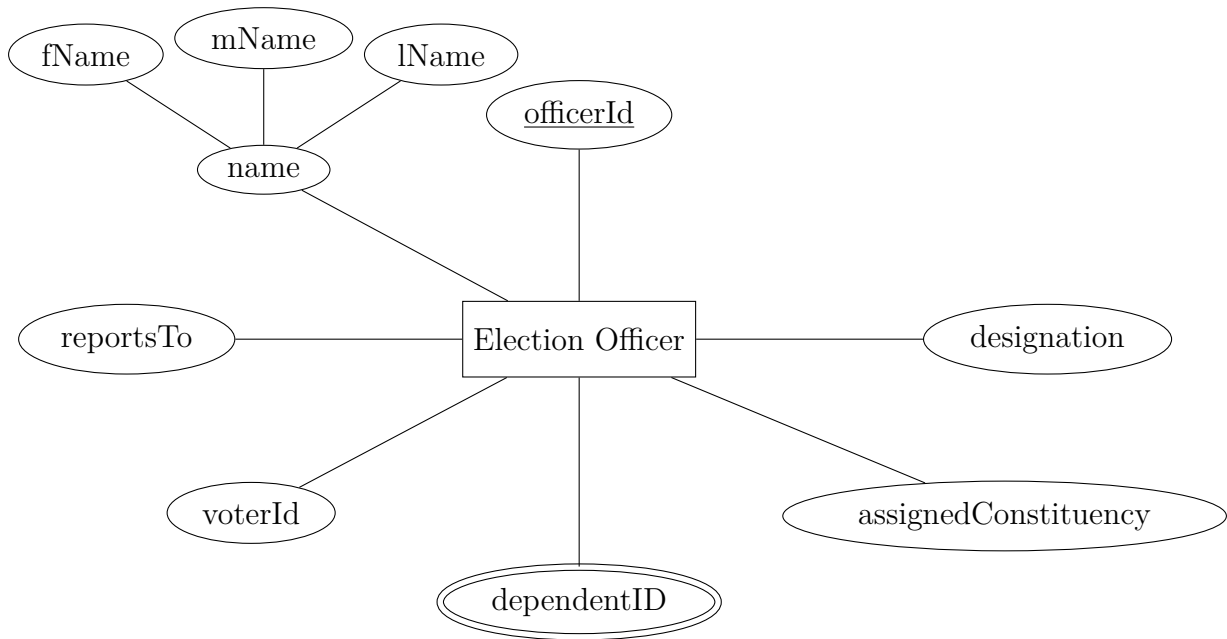


3.7 Election Officer

Represents officials appointed by the Election Commission to supervise and manage the election process. Election officers are responsible for ensuring that voting is conducted fairly and smoothly in their assigned constituencies. They may supervise polling booths, manage election logistics, and report to higher authorities within the election administration.

- **officerId** : Uniquely identifies each election officer and acts as the primary key.
- **name** : A composite attribute consisting of first, middle and last name.
- **designation** : Specifies the role or designation of the officer within the election administration.

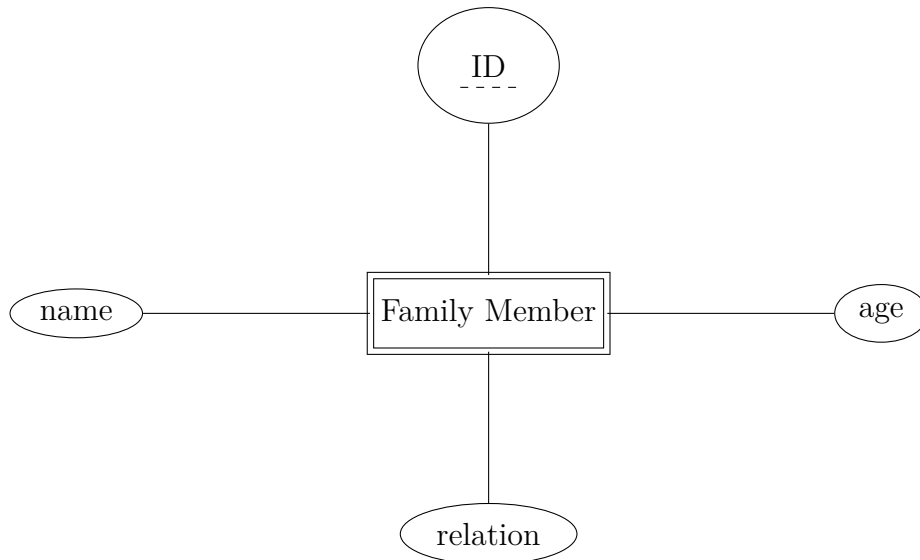
- **reportsTo** : Stores the identifier of the superior officer to whom the officer reports.
- **dependentID** : Represents dependent family members associated with the officer for administrative records.
- **voterId** : Identifies the voter record corresponding to the officer.
- **assignedConstituency** : Identifies the constituency where the officer is assigned for election duty.



3.8 Family Member

Represents the dependent family members associated with an individual in the system. These records are maintained for administrative or identification purposes and capture basic personal details along with the relationship to the primary individual.

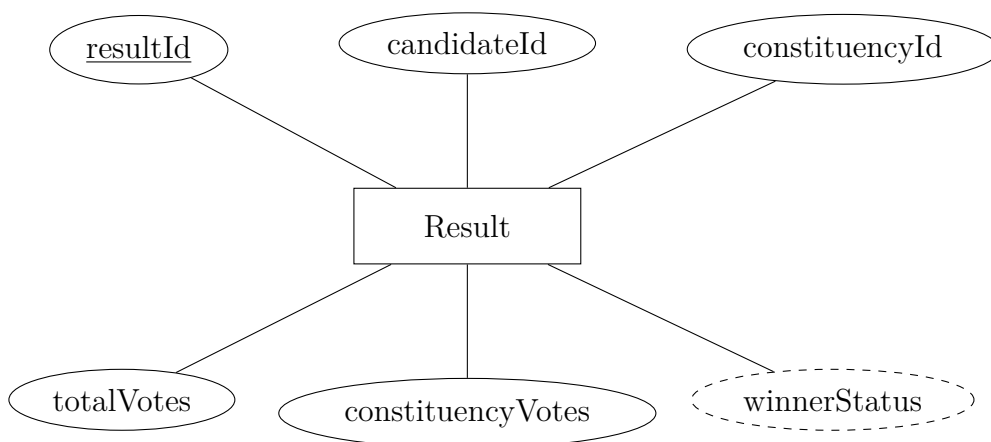
- **ID** : Uniquely identifies each family member and acts as the primary key.
- **name** : Stores the full name of the family member.
- **age** : Represents the age of the family member.
- **relation** : Specifies the relationship of the family member with the primary individual (e.g., spouse, child, parent).



3.9 Result

The Result entity stores the final outcome of an election for each candidate in a constituency. It records the total number of votes received by candidates and identifies whether the candidate won or lost the election in that constituency.

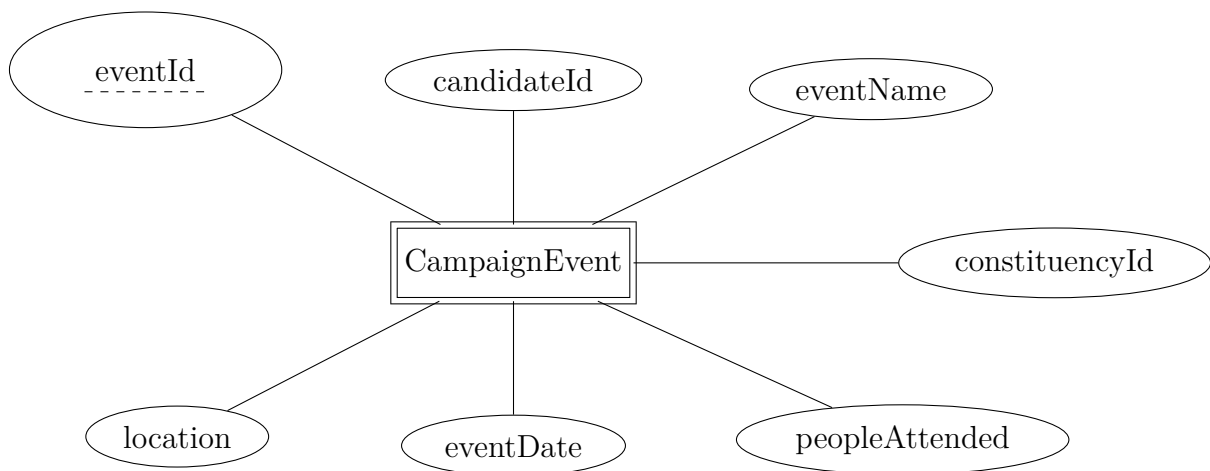
- **resultId** : Primary key that uniquely identifies each result record.
- **candidateId** : References the candidate whose vote count is being recorded.
- **constituencyId** : Identifies the constituency where the result is declared.
- **totalVotes** : Total number of votes received by the candidate in that constituency.
- **winnerStatus** : Indicates whether the candidate won the election (Winner / Not Winner).
- **constituencyVotes** : Number of votes cast in that constituency.



3.10 Campaign Event

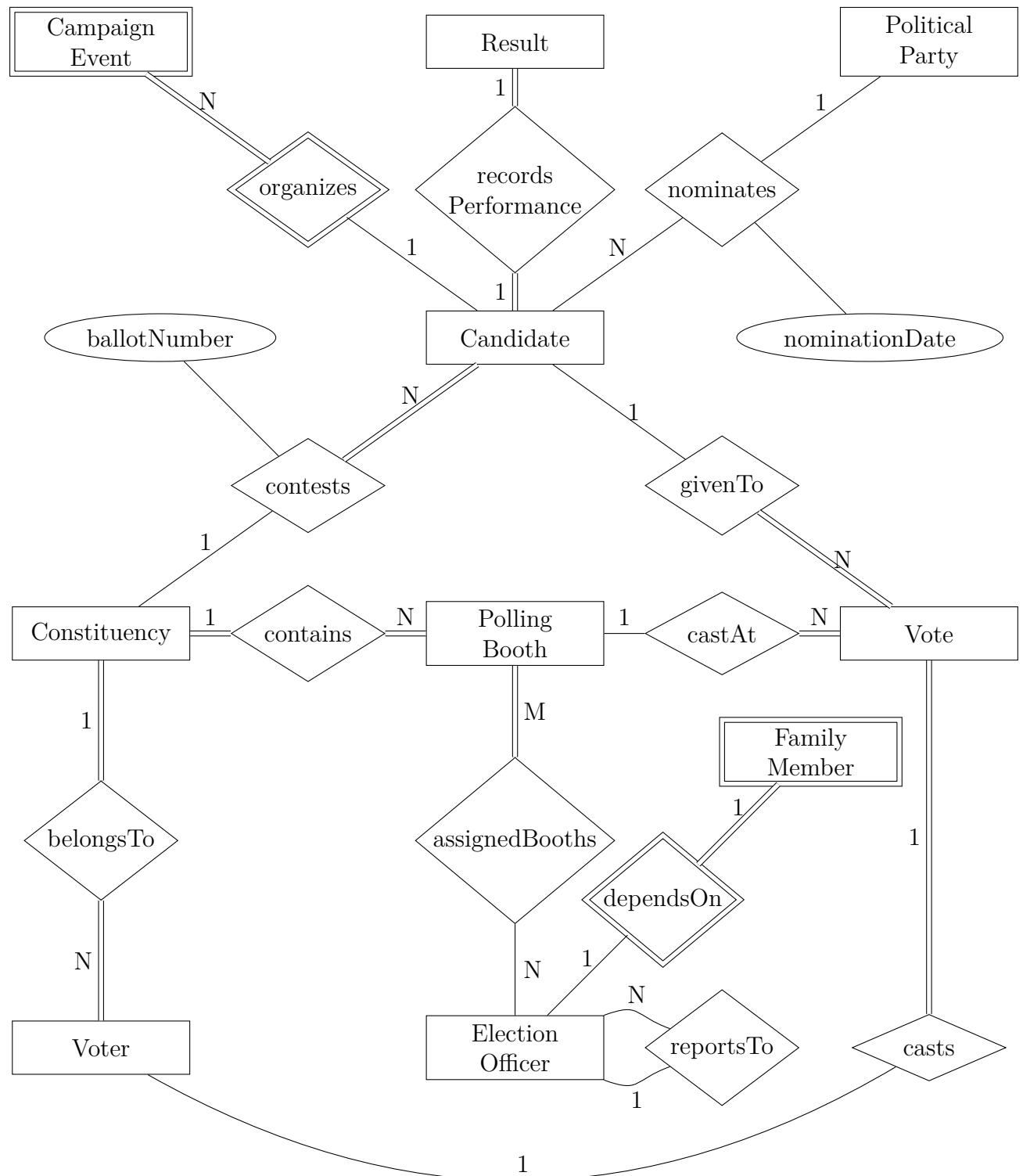
Campaign events represent political rallies, meetings, or public gatherings organized by candidates during the election period. These events help candidates interact with voters, present their policies, and gain support within a constituency. Each event is associated with a specific candidate and takes place at a particular location on a scheduled date.

- **eventId** : Partial key that uniquely identifies each campaign event when candidateId is given.
- **candidateId** : Identifies the candidate who organized the campaign event.
- **eventName** : Name or title of the campaign event.
- **location** : Venue where the campaign event is conducted.
- **constituencyId** : Constituency where the event takes place.
- **eventDate** : Date on which the campaign event is held.
- **peopleAttended** : Number of people who attended the campaign event.



4 ER Diagram

The ER diagram represents the overall structure of the election database and shows how different components of the system interact with each other.



5 Relationships

The election database includes several interactions between the different components of the system.

5.1 Candidate Contests in Constituency

Each candidate contests the election from exactly one constituency in a particular election. However, a constituency may have multiple candidates contesting the election.

The **ballotNumber** indicates the position assigned to a candidate on the ballot paper for that particular constituency.

Relationship type : Binary

Entity 1 : Candidate

Entity 2 : Constituency

Relationship : contests

Cardinality: Many-to-One (N:1)

- One constituency can have many candidates.
- Every candidate will definitely participate in elections for exactly one constituency.

5.2 Political Party Nominates Candidate

Political parties nominate candidates to contest elections in different constituencies. A political party may nominate multiple candidates. However, a candidate usually belongs to only one political party. Independent candidates may not belong to any party.

The **nominationDate** attribute records the date on which a political party officially nominates a candidate for the election.

Relationship type : Binary

Entity 1 : Political Party

Entity 2 : Candidate

Relationship : nominates

Cardinality: One-to-Many (1:N)

- One political party can nominate many candidates.
- A candidate belongs to at most one political party.

5.3 Voter Belongs to Constituency

Each voter is registered in a particular constituency based on their place of residence. A constituency contains many registered voters.

Relationship type : Binary

Entity 1 : Voter

Entity 2 : Constituency

Relationship : belongsTo

Cardinality: Many-to-One (N:1)

- Each constituency must have atleast one voter.
- Each voter belongs to exactly one constituency.

5.4 Constituency Contains Polling Booths

Polling booths are set up within constituencies to allow voters to cast their votes. Each constituency contains several polling booths, but every polling booth belongs to only one constituency.

Relationship type : Binary

Entity 1 : Constituency

Entity 2 : Polling Booth

Relationship : contains

Cardinality: One-to-Many (1:N)

- Each constituency contains atleast one polling booth.
- Each polling booth belongs to one constituency.

5.5 Election Officers Assigned to Booths

Officers are assigned to polling booths to manage the voting process and ensure smooth conduct of the election.

Relationship type : Binary

Entity 1 : Election Officer

Entity 2 : Polling Booth

Relationship : assignedToBooths

Cardinality: Many-to-Many (M:N)

- One officer can be assigned to multiple booths.
- A booth must have atleast one supervising officer assigned to it.

5.6 Voter Casts Vote

During an election, each voter can cast one vote. The vote record stores information about the voter who cast the vote.

Relationship type : Binary

Entity 1 : Voter

Entity 2 : Vote

Relationship : casts

Cardinality: One-to-One (1:1)

- Each voter can cast only one vote in an election.
- Each vote is associated with exactly one voter.

5.7 Vote Given to Candidate

Each vote is cast in favor of a particular candidate. A candidate may receive many votes from different voters.

Relationship type : Binary

Entity 1 : Vote

Entity 2 : Candidate

Relationship : givenTo

Cardinality: Many-to-One (N:1)

- One candidate can receive many votes.
- Each vote is cast for exactly one candidate.

5.8 Candidate Organizes Campaign Events

Candidates organize campaign events such as rallies or meetings to interact with voters and promote their ideas.

Events can't be conducted without a candidate. Hence campaign events will be a weak entity and relationship between candidate and campaign event will be identifying relationship, with candidate as owner entity.

Relationship type : Identifying

Entity 1 : Candidate

Entity 2 : Campaign Events

Relationship : organizes

Cardinality: One-to-Many (1:N)

- One candidate can organize multiple campaign events.
- Each campaign event is organized by one candidate.

5.9 Result Records Candidate Performance

The result entity records the number of votes received by each candidate in a constituency and determines whether the candidate won the election.

Relationship type : Binary

Entity 1 : Result

Entity 2 : Candidate

Relationship : recordsPerformance

Cardinality: One-to-One (1:1)

- Each candidate has one result record for the election.
- Each result corresponds to one candidate.

5.10 Vote Cast at Polling Booth

Votes are cast at polling booths during the election process. Each vote is recorded at the polling booth where it was cast.

Relationship type : Binary

Entity 1 : Vote

Entity 2 : Polling Booth

Relationship : castAt

Cardinality: One-to-Many (1:N)

- One polling booth records many votes.
- Each vote is cast at exactly one polling booth.

5.11 ElectionOfficer reportsTo Election Officer

Every election officer will have an officer whom he reports to.

Relationship type : Recursive Binary

Entity 1 : Election Officer

Entity 2 : Election Officer

Relationship : reportsTo

Cardinality: Many-to-One (N:1)

- One Election Officer reports to one Election Officer.
- Many election officers can report to one superior election officer.

5.12 Family Member dependsOn Election Officer

Every election officer will have atmost one dependent who has ID(similar to aadhaar). Voter here acts as weak entity and Election Officer acts as owner entity.

We record the family members data only when Election Officer exists. Hence Election Officer acts as owner entity, Family Member is weak entity.

Relationship type : Identifying Relationship

Entity 1 : Family Member

Entity 2 : Election Officer

Relationship : dependsOn

Cardinality: One-to-One (1:1)

- One Election Officer has one family member who is dependent on him/her.
- One member can depend only one one of the Election Officer.

6 Relational Schema with Relational Integrity Constraints (RICs)

PoliticalParty(partyId, Name, partySymbol, partyLeaderId, foundedYear, partyFounderId, headquarters)

Result(resultId, candidateId, constituencyId, totalVotes, constituencyVotes, winnerStatus)

CampaignEvent(eventId, candidateId, eventName, location, constituencyId, eventDate, peopleAttended)

Candidate(candidateId, Name, dateOfBirth, age, gender, partyId, constituencyId, phoneNumber)

Constituency(constituencyId, constituencyName, areaSqKm, totalVoters, numberOfPollingBooths, districtName)

Voter(voterId, Name, dateOfBirth, age, gender, constituencyId, phoneNumbers)

Vote(voteId, voterId, candidateId, boothId, electionId, timestamp)

PollingBooth(boothId, location, constituencyId, capacity, boothNumber)

ElectionOfficer(officerId, name, designation, reportsTo, voterId, dependentID, assignedConstituency)

FamilyMember(ID, name, age, relation, officerId)